

Ali Mehdi Khan

Lucknow, Uttar Pradesh, India | +91-9569042552 | ali973mehdi@gmail.com
linkedin.com/in/ali-mehdi-khan-b4062b2a3 | github.com/alimehdikhan | alimehdikhan.github.io

Professional Summary

B.Tech Computer Science graduate (2026) with hands-on experience building production-grade LLM-powered applications, NLP pipelines, and backend REST APIs using Python and FastAPI. Engineered a full-stack AI system integrating OpenAI Whisper for real-time speech transcription and phoneme scoring. Experienced in prompt engineering, LLM evaluation, and deploying AI services to the cloud. Knowledgeable in agentic AI architectures (LangChain, LangGraph), RAG pipelines, and vector databases. Google Cloud certified in Vertex AI, Gemini, and Imagen.

Technical Skills

Languages: Python, Java, JavaScript, SQL, C, C++

AI / LLM: LLMs (OpenAI API, Whisper), Prompt Engineering, LLM Evaluation, NLP, TensorFlow, Keras

Agentic AI & RAG: LangChain, LangGraph, RAG Pipelines, Vector Databases (FAISS; Pinecone, Weaviate)

Backend & APIs: FastAPI, REST APIs, Next.js, SvelteKit, Hono, HTML, CSS

Cloud & DevOps: Google Cloud (Vertex AI, Gemini, Imagen), Hugging Face Spaces, Docker, Git, GitHub

Experience

Junior Developer — Web & Mobile Applications

Aug 2026 – Present

ImaPro

Lucknow, India

- Modernized the production frontend by migrating it to SvelteKit 5, reducing component re-render overhead and improving page load responsiveness, by restructuring the UI into reactive, reusable components with SvelteKit's updated state and props model.
- Maintained zero regressions in core booking and data flows during the migration, verified through manual QA across key user journeys, by preserving the existing Hono/better-sqlite3 backend contract and validating each migrated route against prior behavior.
- Improved cross-device usability for web and mobile users, measured by consistent UI behavior across breakpoints, by implementing responsive layouts and shared component patterns across the application.
- Accelerated team onboarding and code review turnaround, evidenced by faster PR cycle times, by documenting component structure and contributing to internal engineering conventions for the new frontend stack.

Machine Learning Intern

June 2025 – July 2025

GrasTech

Lucknow, India

- Developed supervised and deep learning models for skin cancer detection and diabetes prediction in Python (TensorFlow/Keras), applying feature engineering and hyperparameter tuning to maximize predictive accuracy on real-world medical datasets.

Projects

AI Pronunciation Coach / Python, FastAPI, OpenAI Whisper, NLP, Epitran, Docker, Hugging Face

- Architected an end-to-end LLM-powered pronunciation coaching platform integrating OpenAI Whisper for real-time ASR transcription and deep phoneme-level spoken English analysis.
- Engineered a low-latency audio processing and scoring pipeline using FastAPI; designed prompt-driven LLM feedback loops to generate actionable, user-facing coaching insights with sub-second response times.
- Implemented a custom NLP engine with Epitran for IPA phoneme extraction and word alignment, powering a transparent per-word pronunciation scoring system.
- Containerized the stack with Docker and deployed to Hugging Face Spaces for cloud-based, reproducible inference via a public API endpoint.

Cancer Detection System / Python, Keras, TensorFlow, CNN, Deep Learning

- Trained a binary CNN classifier on real-world medical imaging datasets using Keras deep learning architectures, achieving 90%+ accuracy for early-stage cancer prediction.
- Optimized model generalization through systematic hyperparameter search and data augmentation pipelines, reducing overfitting and improving F1-score on held-out validation sets.

Education

Bachelor of Technology – Computer Science and Engineering

2022 – 2026

Babu Banarasi Das University (BBDU)

Lucknow, India

Certifications

- Build Real-World AI Apps with Gemini & Imagen – Google Cloud Skill Badge, 2025
- Prompt Design in Vertex AI – Google Cloud Skill Badge, 2025
- Machine Learning with Python – freeCodeCamp, July 2025
- Technology Job Simulation – Deloitte via Forage, July 2025

Volunteering

Programming Instructor – Tech for Good

2025

Volunteer

Lucknow, India

- Delivered programming training to 30+ underprivileged students over 3 months, enabling participants to build their first functional applications.